

ESD Guidance

Water Sensitive Urban Design

What is water sensitive urban design

Designing urban environments with water sensitive urban design (WSUD) views rainwater as valued resource, encourages stormwater to permeate into the ground where possible, and cleans and slows any excess stormwater to reduce the harm it causes to our creeks and rivers.



Why is WSUD important

In undeveloped land, rainfall is free to soak into the ground which recharges aquifers, flows across the land which irrigates vegetation, and evaporation, and plant transpiration, puts moisture into the air – feeding the rain cycle.

With urban development comes more roofs, roads, and other paving, and a corresponding reduction in permeable ground area. Increases in impermeable surface area magnify the volume and speed of stormwater entering the drainage system, and the pollutants that come with it.

Rather than the benefitting the environment, the high volumes of fast-flowing, and polluted stormwater disturb the ecosystem. Water sensitive urban design helps reduce these negative impacts.

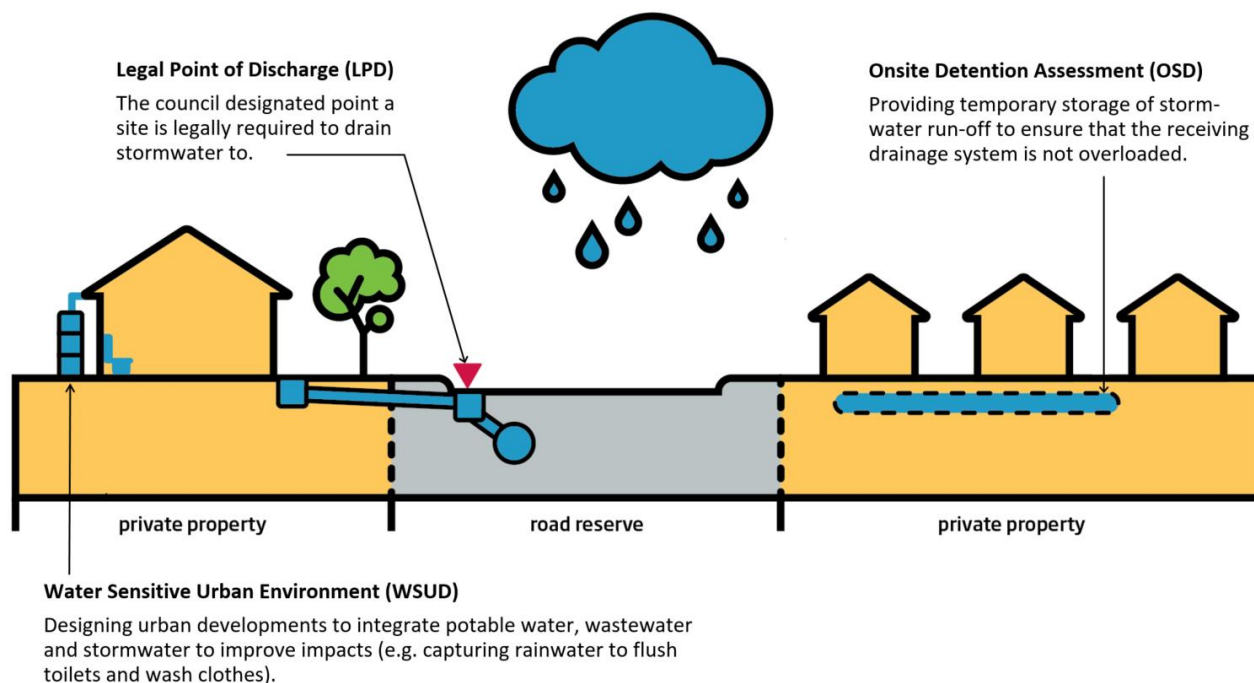
For more information see Melbourne Water's [Introduction to WSUD](#).

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Stormwater terms in Brimbank

Stormwater management in new development involves several Brimbank departments. There are also a number of different terms used that relate to different aspects of stormwater management. Stormwater 'drainage' is distinct from 'sanitary drainage' (sewers).



Water Sensitive Urban Design (WSUD) and Integrated Water Management (IWM)

- Integrated Water Management coordinates development and management of water, land and related resources, and Water Sensitive Urban Design is the application of this in the design of urban environments.
- WSUD 'best practice' is reviewed as part of the assessment of [planning applications](#) within the City Planning Department.

Stormwater drainage

- On-site drainage is designed to allow stormwater to safely drain to the stormwater drainage system, and avoid flooding.
- [Drainage design applications](#) are assessed by the Brimbank Engineering Services Department.

On-site Detention (OSD)

- Larger developments are required to have stormwater systems designed to temporarily 'detain' stormwater on-site. On-site Detention (OSD) systems allow stormwater to drain slowly after heavy rainfall to avoid overloading the drainage system.
- Brimbank Engineering Services provide [Property drainage design and onsite detention assessment guidance](#) including specific information for the design of OSD systems.
- OSD is assessed by the Engineering Services Department. A fee applies.

Legal point of discharge (LPD)

- The legal point of discharge (LPD) is the Council designated point a site is legally required to drain stormwater to.
- Apply to find out the LPD for your site on the Engineering Services [Property Information](#) page. A fee applies.

WSUD in planning applications

Improving stormwater outcomes in new development

Both [Clause 22.02](#) and [Clause 53.18](#) of the Brimbank Planning Scheme call for best practice in stormwater management. These policies have a range of objectives that focus on improving urban stormwater outcomes in the design, construction and the day to day operation of new development.

These objectives are listed under the Water and Stormwater headings on [The 10 Sustainable Building categories](#), along with useful resources and information on how best practice in stormwater management is to be demonstrated.

This Guide (ESD Guidance: Water Sensitive Urban Design) sets out preferred treatments methods and strategies for development in Brimbank including:

- a clear hierarchy of off and on-site treatment strategies
- encouraged WSUD treatment methods
- non-supported WSUD treatment methods
- preferred WSUD strategies for particular building genres
- and what WSUD information to submit with your planning application.

What type of WSUD treatment is sought

Finding an appropriate WSUD strategy for a development will depend on the type of development, the climate and the location. Brimbank is built on the flat, volcanic, basalt plains, that extend to west of Melbourne, has relatively lower rainfall, and soils are generally more clay-like.

Applicants are asked to observe a treatment hierarchy when designing WSUD strategies for development in Brimbank. The Brimbank WSUD Hierarchy asks for:

- the area of stormwater catchments to be minimised
- stormwater to be treated on-site wherever possible
- off-site treatment only to be sought once all on-site treatment options have been exhausted.

BRIMBANK WSUD HIERARCHY	
MINIMISE CATCHMENTS	1. Minimise paving extent 2. Utilise permeable paving in pedestrian areas* * (permeable paving is generally not supported in areas subject to vehicle movement)
TREAT ON-SITE	3. Capture rainwater for reuse 4. Treat with bio-filtration 5. Treat with proprietary products* * (bio-filtration options need to be demonstrated exhausted before proprietary products are used)
TREAT OFF-SITE	6. Drainage schemes * (on-site options need to be demonstrated exhausted before drainage schemes are considered)

What WSUD information do I need to submit

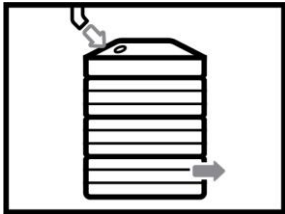
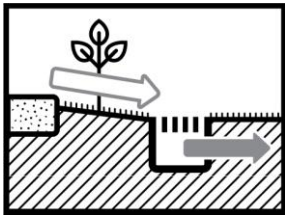
When making development applications, applicants are asked to demonstrate best practice in stormwater management through the submission of stormwater modelling, WSUD plans and other details. These are summarised on the Environmentally Sustainable Design (ESD) pages of the City Planning website under the [What ESD information do I need to submit](#) heading, though detail is provided below.

Where an ESD report is required WSUD initiatives should be incorporated within the Water and Stormwater categories rather than a separate report. Where applications do not require an ESD submission, best practice in stormwater management should be demonstrated in a WSUD report.

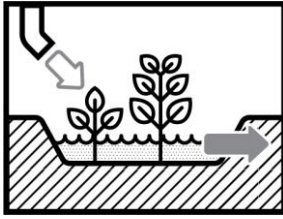
Provide the following with your planning application:	
DESIGN	• WSUD strategy – WSUD strategies are expected to: - generally follow the <i>Brimbank WSUD hierarchy</i> above - select treatments from the <i>Encouraged WSUD treatment methods</i> listed below - avoid treatment from the <i>Non-supported methods WSUD treatment methods</i> also listed below. • Modelling: - STORM – provide a separate line for the treated and non-treated portion of each building's roofs, terraces, impermeable paving, etc.), and demonstrate a minimum 100% rating - MUSIC – Provide a modelling report which demonstrates the best practice minimum indicators have been achieved, and provide an electronic file ('sqr' format) for review. • WSUD plan: - show the full site with the treated and non-treated portion of all new and altered catchments - indicate the area of each per the modelling report - locate treatment devices and annotate with a description (e.g. buffer-strip area, raingarden area and detention depth, tank capacity and reuse connections) per the WSUD modelling. • Treatment device details - Provide typical details of treatment devices appropriate to assess their effectiveness.
CONSTRUCTION	• Provide detail of, or reference to, appropriate erosion and sediment control measures to be employed during construction (this may be included in a Construction Management Plan for larger projects).
OPERATION	• Provide details of maintenance commitments for the treatment devices to ensure their on-going effectiveness for the life of the development. Include inspection frequency, cleanout procedures, and remediation requirements at a minimum.

Encouraged WSUD treatment methods

Listed in order of preference

DEVICE	GENERAL	NOTES
Rainwater tanks 	• Capture as much roof water as possible for on-site use • Non-potable uses include: toilets, laundries (washing machine cold taps), bin or vehicle wash-down, and irrigation • As irrigation demand is seasonal, more year-round demands, such as toilet flushing, should be prioritised • Runoff from terraces should be filtered before storage, and treated before use.	• Use leaf-guards, first-flush diverters and in-line filters to reduce contaminants entering rainwater tanks • When modelling stormwater demand: - Residential – in STORM use the number of bedrooms in each dwelling - Clothes-washing - to account for connecting washing machine cold taps - claim an extra 'bedroom' per dwelling in STORM - Commercial – if the occupancy is not known use Table D2D18 in NCC 2022 Volume 1 - Industrial – unless occupancy is known allow 1.5 occupants per car parking bay - Reliability - target a tank reliability of 80% in STORM to minimise how often untreated rainwater will overflows from tanks • When capturing runoff from trafficable areas detail the rainwater filtering before storage, and treatment before use.
Buffer strips 	• Buffer-strips direct runoff across either lawn or garden which allows runoff to be absorbed and some pollutants to settle before collection • Only simple, shallow drains are required, which connect to the internal drainage system • Buffer-strips are much simpler to construct and maintain than more complex treatments such as raingardens • The use of gardens as a buffer is preferred lawn – to minimise the pedestrians stepping on sodden ground in wet weather.	• Indicate the area of paving to be treated by any buffer strip, and show indicative falls to it. • Indicate and annotate the area of vegetation • Note a minimum runoff travel distance of 0.6 m across vegetation • Show a surface drain, and note it is to be connected to the internal drainage system • Detail the paving and buffer-strip to ensure there is adequate fall to avoid ponding runoff causing reactive soils to heave • Position away from side boundaries so when overwhelmed, runoff will not cross the boundaries • Fall away from buildings to avoid any pooling runoff which could affect walls or footings.

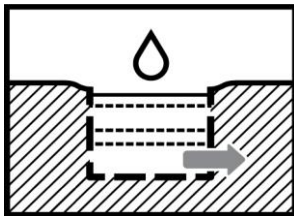
Vegetated swales



- Vegetated swales are artificial mini-creeks that allow stormwater absorption, which provides passive irrigation for surrounding vegetation
- Excess flow is slowed and cleaned by the vegetation which removes sediment, nutrients and catches 'gross pollutants' (e.g. drink bottles, plastic bags, etc.)
- At the end of the swale a simple surface drain collects excess stormwater and channels this to the stormwater drainage system
- Swales are simpler to construct than raingardens and offer urban greening and cooling.

- Show swale routes on plans, and annotate their area
- Indicate a surface drain at the end, and note it is connected to internal drainage system
- Provide a typical detail which notes basic dimensions, planting species and the number of plants per m²
- Provide a maintenance regime to ensure on-going effectiveness
- Vegetated swales can be entered in STORM modelling as a 'buffer-strip', however, model in MUSIC to gain a greater credit for the treatment provided.

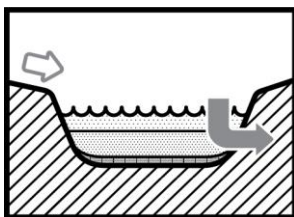
Raingardens



- Avoid raingardens wherever possible:
 - reduce catchment area
 - optimise rainwater capture
 - utilise vegetated buffer-strips or other more intensely vegetated swales as these are simpler and provide better passive irrigation
- Where unavoidable minimise the area and number of raingardens where possible
- Line in-ground raingardens to prevent damage to adjacent structures
- Consider raingardens that have a submerged zone at the base, as the retained moisture help plants survive long dry spells
- Periodic full replacement of the filter media is required when the media becomes sediment laden over time (e.g. every 1–20 years).

- Where unavoidable:
 - Townhouses - provide a minimal number, and use to treat shared runoff (e.g. common driveways) so owner's corporations rather than individual owners are responsible for their maintenance
 - Larger projects – distribute raingardens to limit the distance runoff needs to travel
- Locate raingardens so that when overwhelmed, runoff will not cross side boundaries, or pool near buildings
- Position in high-traffic areas to increase visibility and encourage maintenance
- On WSUD plans annotate the area of each raingarden, its detention depth, indicate the paving area to be treated and show indicative falls
- Provide typical details: basic dimensions, filter media, lining, number of plants per m², and species, scouring prevention, etc.
- Provide a maintenance regime to ensure on-going effectiveness.
- Confirm in-ground raingardens can drain via gravity to the legal point of discharge wherever possible.

Sediment basins

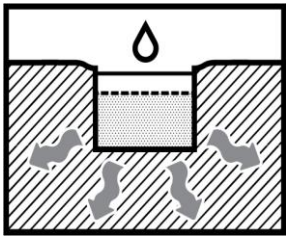
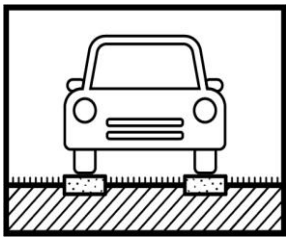
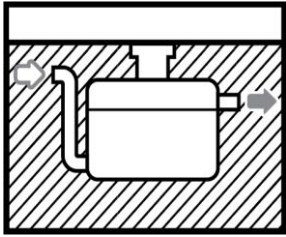


- Sediment basins are used with large developments (e.g. subdivisions) due to their ability to process high volumes of stormwater
- They can be used in series with other treatment devices as part of a 'treatment train'.

- Provide a typical detail with: basic dimensions, filter media, any lining, planting species, planting density, and scouring prevention
- Submit a MUSIC modelling report, as well as the electronic file (in '.sqr' format).

Non-supported WSUD treatment methods

Noting exception criteria

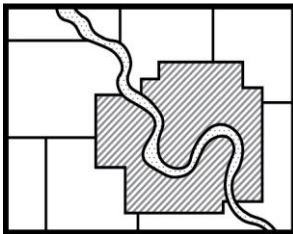
DEVICE	GENERAL	NOTES
Infiltration trenches 	- Water doesn't flow through clay soils (infiltration) and these are common in Brimbank - Due to this, infiltration trenches are not considered feasible in Brimbank.	- Exceptions will only be considered where: - it can be demonstrated other treatment methods are inappropriate and on-site soil hydrology allows appropriate infiltration - an engineered solution is provided - any trench is away from infrastructure which it might have a detrimental effect on.
Split-driveways 	- The use of 'tyre-track' style driveways is not supported as the central vegetated strips are too readily damaged when driven on - This results in exposed earth, and then mud in wet weather.	Exceptions are not generally considered.
Proprietary devices 	- The use of proprietary devices is not supported until preferred treatment methods are exhausted. - Concerns for proprietary devices are that generally: - they do not support urban greening or cooling objectives - they do not allow stormwater to infiltrate into the landscape - industry protocols for testing and certification of individual devices haven't been fully applied - the lack of visibility of buried devices increases the likelihood their maintenance is forgotten - the risk supply of parts may not be available for the device for life of the development they serve.	- Alternatives exhausted - To be considered, the use of alternate preferred treatment devices is to be demonstrated as being exhausted, or not appropriate - Performance - Evidence of device performance is to be provided from an appropriately experienced and reputable third-party reviewer - Parts - The supply of the device and required replacement parts is to be available in Melbourne for the foreseeable future - Maintenance - The maintenance regime required to maintain the on-going effectiveness of the device is committed to by the applicant for the life of the development - Ownership - Splitting development into multiple, individual lots lessens the likelihood of effective professional maintenance.

Permeable paving



- Porous paving is encouraged in pedestrian areas
- Use in areas subject to vehicle movement is supported areas required by an arborist; and portions of commercial/industrial car parking bays
- If substrate design doesn't allow runoff to readily drain away the build-up of moisture can lead to the soil below to heave
- Bucking paving causes trip-hazards, and subsidence which allows ponding in wet weather
- It is important permeable paving is:
 - laid on an appropriate substrate
 - that this substrate is drained so it doesn't allow water to pool and cause ground heave
 - regularly maintained to ensure the paving remains porous as designed.
- Use will be considered where:
 - Houses – For attached and detached housing (class 1) where required by arborist for tree health
 - Non-residential – in portions of car parking bays where torque from turning car wheels is not likely to cause pavers to move (e.g. at the nose of parking bay)
- If proposed, demonstrate:
 - Performance – evidence of tested performance (e.g. noting vehicle type and movement)
 - Detail – appropriate engineered details for review by the Brimbank's Traffic team (e.g. a cross-section with appropriate substrate with sub-surface drainage)
 - Maintenance – A commitment to a maintenance regime that ensures the on-going effectiveness of the device for the life of the project
 - Monolithic slabs - Permeable monolithic slabs are preferred over separate 'unitised' pavers, though when used acknowledgement is sought that the applicant is prepared to commit to the increased construction cost.

Drainage schemes



- The use of off-site treatment is not generally supported as planning policy places the onus on applicants to provide best practice treatment onsite.
- Exceptions may be considered where:
 - the use of on-site treatment devices has been exhausted
 - off-site treatment is by a recognised stormwater drainage scheme
 - payment of a contribution to such a scheme is evidenced, once approved by the responsible authority.

WSUD within landscaped areas

The large areas of paving associated with industrial and commercial developments produce high volumes of stormwater runoff.

The more contact stormwater has with landscaping, the greater the potential for passive irrigation. This supports plant growth, natural habitats, and 'urban cooling' per the planning objectives of the Brimbank Planning Scheme.

Designers are asked to consider incorporating bio-filtration, stormwater treatment into their landscape design. This includes devices such as tree pits, vegetated swales and raingardens which all utilise vegetation to treat stormwater.

Passive irrigation

Garden beds, lawn and tree pits can receive stormwater runoff from paved areas when designed with drains connected to the internal drainage system to drain excess runoff:

- providing passive irrigation for landscaping is strongly valued
- tree pit design should include appropriate drainage to avoid detrimental water-logging
- excess stormwater may require additional treatment.

Vegetated swales

Vegetated swales are artificial mini-creeks created to treat stormwater as it flows to a drainage point:

- swales are simple to construct, and can be designed to enhance landscaped buffers
- the ground absorbs some runoff as it flows along the swale, providing passive irrigation for vegetation
- some sediment falls out of the runoff as flow is slowed, and nutrients absorbed by vegetation
- as swales provide less treatment per square metre than raingardens, they suite larger developments
- To gain the best acknowledgement of the treatment provided modelling is encouraged in the MUSIC rather than the STORM modelling tool.

In-ground raingardens

In-ground raingardens provide a potent way to treat stormwater runoff, though they are complicated to build, require regular maintenance, and periodic replacement to ensure on-going effectiveness.

- Fully-lined – Raingardens are generally sought to be fully-lined with a water impervious barrier to avoid reactive clay soils becoming hydrated and damaging infrastructure as they swell.
- Part-lined – Part-lining may only be required on the side of raingardens close to paving, kerbs, etc. in some circumstances when well away from buildings. Geo-fabric is still generally required on the other sides to contain filter media.
- Non-lined – When isolated in landscape areas, some raingardens might avoid lining.
- Treed raingardens – Raingarden lining will limit the expansion of tree roots, and thus tree growth, so trees are discouraged from being planted within, or immediately adjacent to raingardens. When allowed - greater raingarden depth, deeper drainage, species able to tolerate the likely moisture variance, and resist overturning in high-winds are sought.
- Planting – Raingardens require planting of sufficient density to extract the required nutrients, and species selected that prosper in sand/soil media mix, where moisture can vary between being fully saturated or dry for extended periods.

Raingardens and their variations

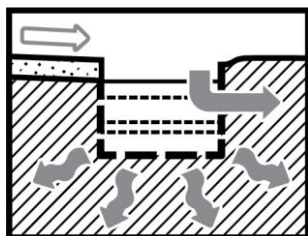
Raingardens are designed to filter pollutants from stormwater which aids the health of receiving water bodies such as rivers and Port Phillip Bay:

- litter – gross pollutants (e.g. cigarette butts) will collect on the surface of the raingarden
- cloudiness – ‘suspended solids’ settle in the filter media to make stormwater less muddy
- nutrients – plant roots absorb nutrients from the stormwater.

Raingardens are designed for light rainfall, so design needs to consider the path runoff will take when they overflow in heavy rainfall.

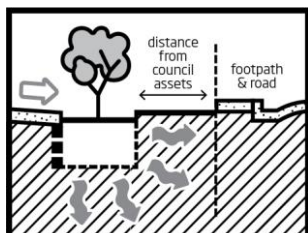
See Melbourne Water’s website for further information on raingardens:

- <https://www.melbournewater.com.au/water-data-and-education/environmental-issues/why-we-need-save-water/tips-saving-water/raingardens>
- <https://www.melbournewater.com.au/building-and-works/stormwater-management/options-treating-stormwater/raingardens>



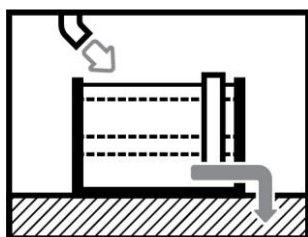
Unlined (infiltration) raingarden

- Known as ‘infiltration’ raingarden, unlined raingardens are designed to allow stormwater to soak into the ground in sandy soils.
- No base drain is needed, just an overflow at the top.
- While useful in sandy soils, they are not-suitable for Brimbank’s more clay-like soils, which don’t allow free drainage.
- See Melbourne Water’s instruction sheet for an example: [Infiltration raingarden instruction sheet](#).



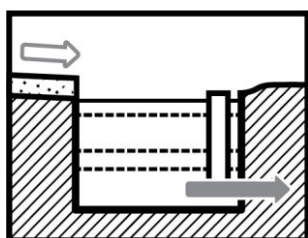
Part-lined raingarden

- Raingardens in close proximity to building are sought to be fully-lined to protect the footings from soil movement.
- Where away from buildings, partial lining can be used to protect adjacent paved car parks, footpaths and roads, yet still allow some infiltration.
- Partial lining also allows the roots of more significant plantings, such as trees to expand into the surrounding soil.



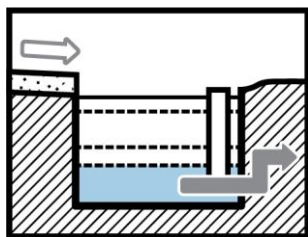
Lined above ground raingarden

- Known as ‘planter box’ raingardens, above ground raingardens position the filtration media above ground.
- These generally can’t be used for treating paving runoff at ground-level, though work well under rainwater downpipes.
- See Melbourne Water’s instruction sheet for an example: [Planter box raingarden instruction sheet](#)



Lined in-ground raingarden

- Fully-lined raingardens are generally sought in Brimbank to avoid stormwater causing the clay soil in the trench walls swelling.
- Generally raingardens are up to 90cm deep, with filtered stormwater drained from the base.
- An overflow drain is provided at the height of the designed ‘detention depth’ above the raingarden, but below surrounding ground level.
- See Melbourne Water’s instruction sheet for an example: [Inground raingarden instruction sheet](#).



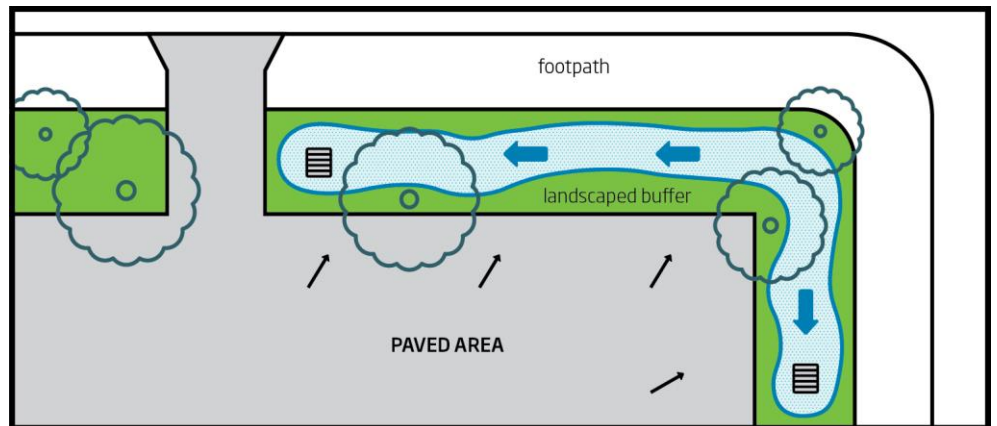
Lined with submerged zone

- To help the vegetation survive in long-dry spells, a variation of the in-ground rain garden is suggested for Melbourne's West.
- Rather than draining from the very base, the outflow drain steps-up, creating a permanently wet 'submerged zone' which plants roots can access
- The shallower depth of the outflow drain requires a less-deep drainage connection.
- See Melbourne Water's instruction sheet for an example: [Raingarden design for Melbourne's west.](#)

Locating bio-filtration in landscaped areas

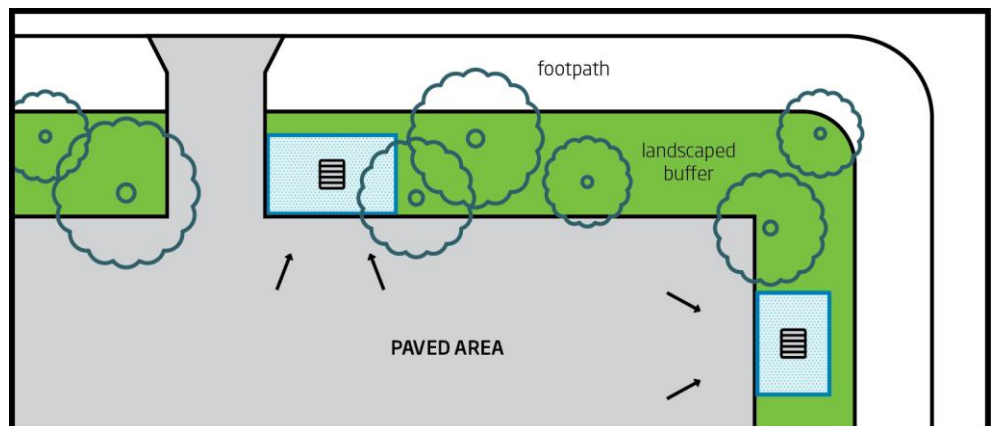
Vegetated swales within landscaped zone

Vegetated swales are shallow, unlined gullies that are simple to construct and offer passive irrigation as they pass adjacent landscaping.



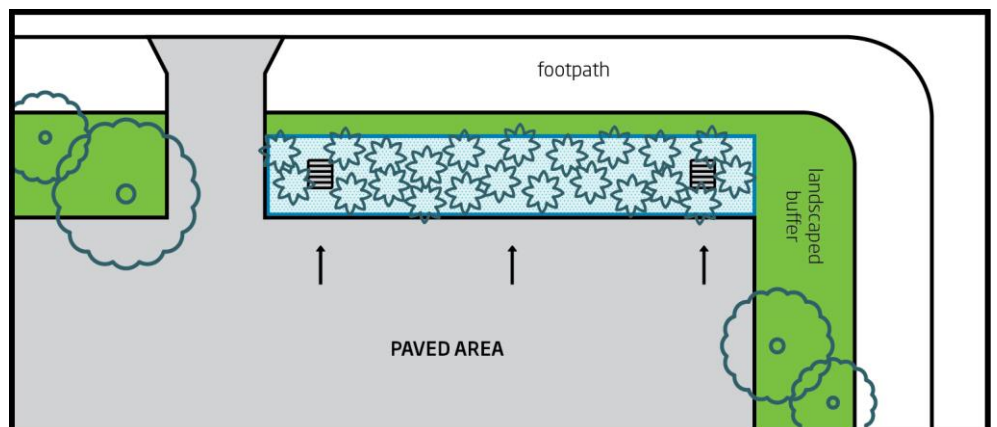
Raingardens interspersed within landscaped zone

Compact raingardens can be distributed among landscaping - allowing space for significant planting, though not passive irrigation.



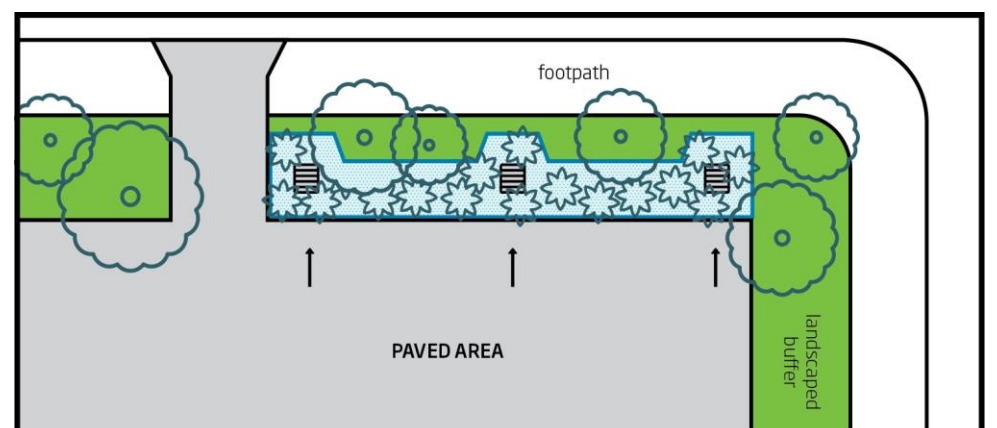
Raingardens displace other vegetation

The lining provided to raingardens limits root expansion, and the size of planting contained.

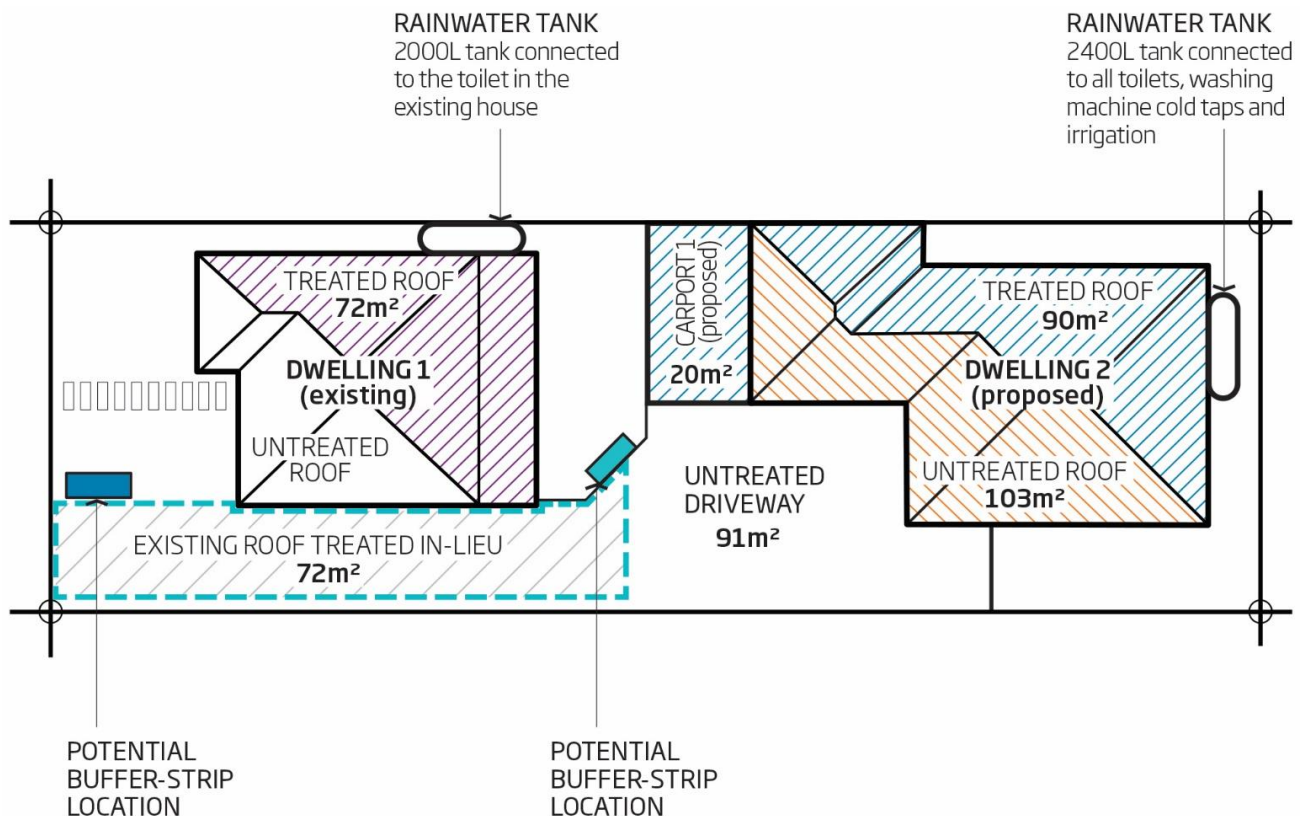


Raingardens aside substantial trees

Lineal raingardens can allow space for significant, in-ground planting though need to be set-back to allow space for tree root expansion.



Preferred strategy: for a new dwelling behind existing



Add a tank to the existing house as well to avoid raingardens on driveways

Design tips

- Typically adding a new dwelling behind an existing results in a long driveway.
- In addition to a tank on the new dwelling, treating the runoff from this driveway commonly requires buffer-strips or raingardens.
- Even though an unaltered existing roof is not required to be treated, if a tank is added – the area of or roof treated can be accepted in-lieu of treating an equivalent area of driveway runoff.
- This 'treat-in-lieu' method is encouraged as it meaningfully utilises additional rainwater while generally avoiding the need for buffer-strips or raingarden.

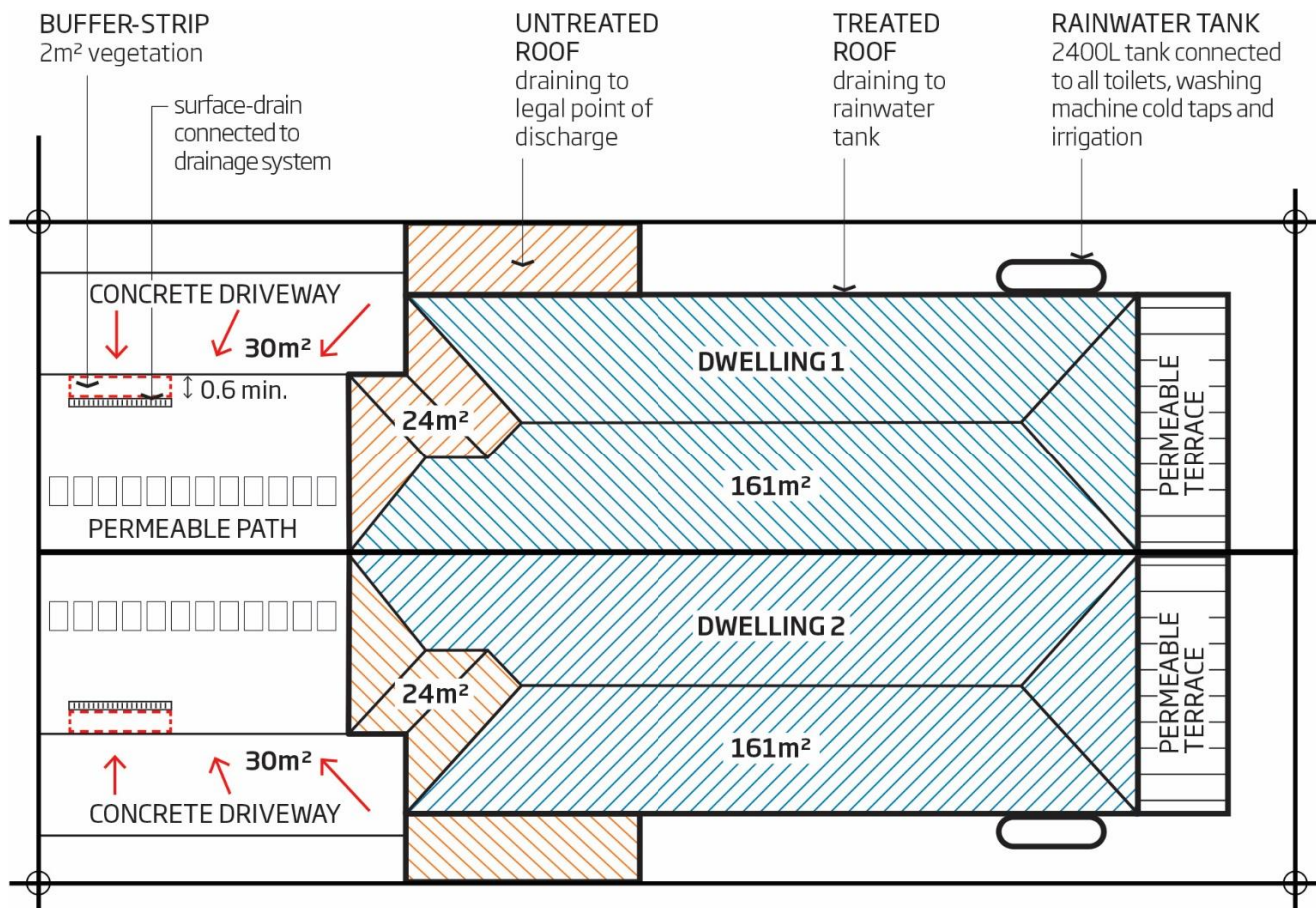
Minimise Catchments

- Minimise catchments for roofs, paths, and driveways
- Use permeable paving in pedestrian areas such as courtyards (not driveways)

Treat On-Site

- Only new and altered catchments need to be included, not existing.
- Tanks – maximise rainwater capture for reuse:
 - Toilets – connect tanks to all toilets in the dwelling they serve
 - Laundry – if connecting cold water washing machine taps to the tank, claim an extra 'bedroom' in STORM to acknowledge the additional demand
 - Treat-in-lieu – If adding a tank to the existing house, claim the area of roof treated in-lieu of new driveways. Enter "untreated driveway" and "treated driveway (existing roof in lieu)" in STORM, and ensure both areas are shown on the WSUD plan.
- Bio-filtration – If additional treatment is needed, simple, vegetated buffer-strips will generally avoid the need for any raingardens.

Preferred strategy: for duplexes



Optimise rainwater use, then simple buffer-strips to avoid raingardens if possible

Design tips

- Position buffer-strips and in-ground raingardens:
 - away from side/rear boundaries to avoid overflowing runoff crossing the boundary
 - away from buildings to avoid pooling water affecting walls or footings.
- WSUD devices are designed to treat lighter rainfall, so think about where runoff will flow when they do overflow.

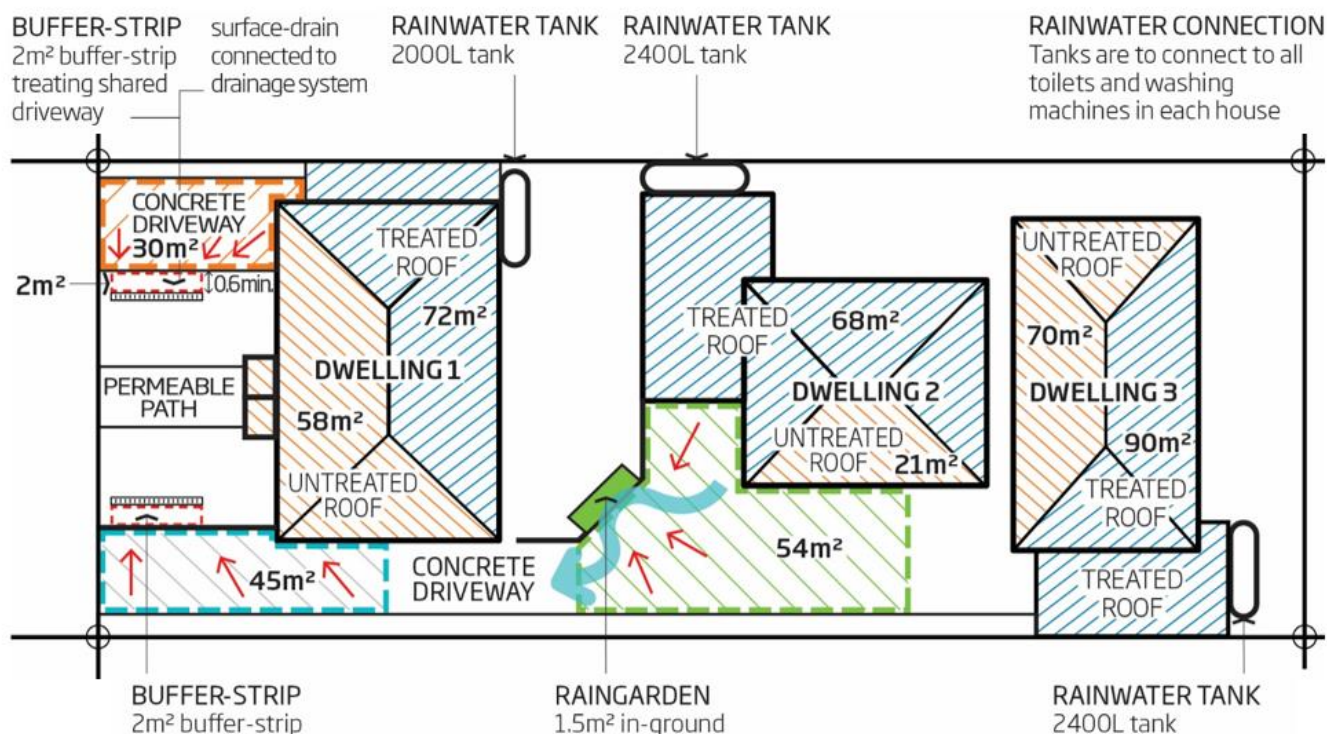
Minimise Catchments

- Minimise catchments for roofs, paths, and driveways
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Treat On-Site

- Tanks – maximise rainwater capture for reuse:
 - toilets – connect tanks to all toilets in the dwelling they serve
 - laundry – when connecting cold water washing machine taps to the tank, claim an extra 'bedroom' in STORM to acknowledge the additional demand
- Bio-filtration:
 - buffer-strips – allow runoff to flow over in-ground vegetation before collecting it in a surface drain
 - Raingardens – when unavoidable, provide locate in highly visible locations at the front of the property to maximise the likelihood maintenance is not forgotten. Use above-ground planter-box raingardens over in-ground raingardens where possible.

Preferred strategy: for townhouses



Optimise rainwater use, then use simple buffer-strips to avoid raingardens

Design tips

- Position raingardens and buffer-strips:
 - away from side/rear boundaries to avoid overflowing runoff crossing the boundary
 - away from buildings to avoid pooling water affecting walls or footings.
- WSUD devices are designed to treat lighter rainfall, so think about where runoff will flow when they do overflow.

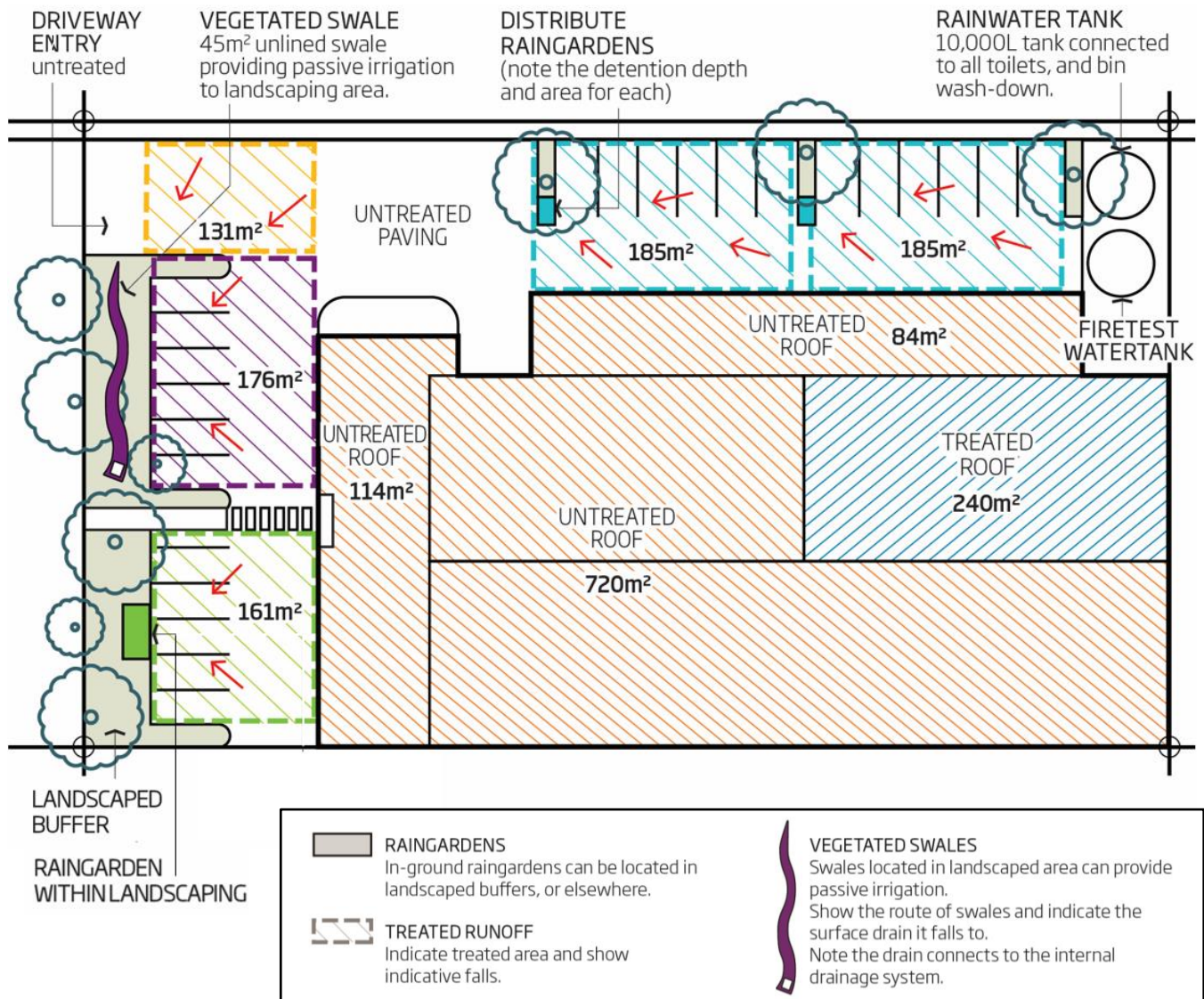
Minimise Catchments

- Minimise catchments for roofs, paths, and driveways
- Use permeable paving in pedestrian areas such as courtyards (not driveways)

Treat On-Site

- Tanks – maximise rainwater capture for reuse:
 - toilets – connect tanks to all toilets in the dwelling they serve
 - laundry – when connecting cold water washing machine taps to the tank, claim an extra 'bedroom' in STORM to acknowledge the additional demand
- Bio-filtration:
 - buffer-strips – allow runoff to flow over in-ground vegetation before collecting it in a surface drain
 - Raingardens – Only use where unavoidable, provide the minimal number possible, locate in shared property to treat shared catchments (e.g. driveways).

Preferred strategy: for industrial sites



Optimise rainwater use, include passive irrigation, and distribute raingardens

Design tips

- Beyond toilet flushing, consider other non-potable water uses (e.g. bin and vehicle wash-down)
- Utilise vegetated swales within landscaped areas to provide passive irrigation and lessen the area of raingardens needed.
- Canopy trees are sought between car bays to shade paving. Though trees aren't supported in raingardens, consider pairing raingardens with tree locations.

Minimise Catchments

- Minimise paving area.
- Use permeable paving in pedestrian areas and at the nose of car bays where there is no torque from turning vehicle wheels (and no heavy vehicles access).

Treat On-Site

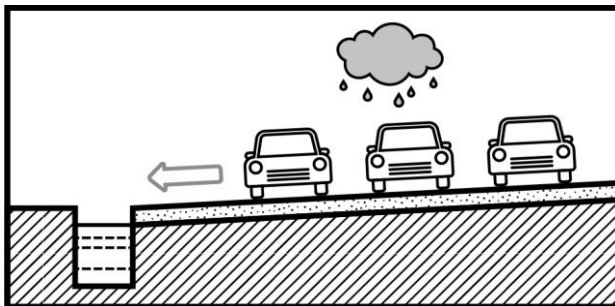
- Tanks – maximise rainwater capture for reuse.
- Bio-filtration – The design is to allow space for sufficient bio-filtration on site. Utilise vegetated swales and other passive irrigation techniques before raingardens.
- Position raingardens:
 - in distributed locations across the site to minimise runoff travel distances
 - off of side/rear boundaries by a distance equivalent to their depth
 - with overflow routes planned - provide kerbs to control any overflow.

Protecting stormwater from paved areas

Most runoff from paved driveways and parking areas will have moderate pollutant loads until there is an incident such as a oil leak from a vehicle, or accidental spills.

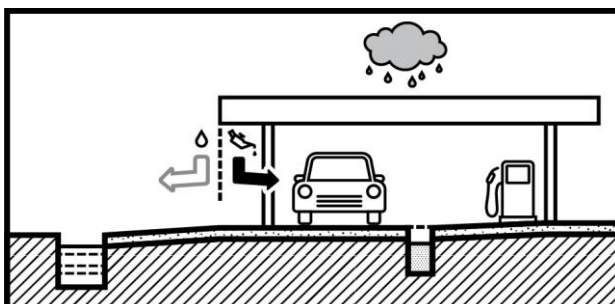
Where industrial uses include handling or processing of goods or materials outdoors, the risk of stormwater becoming polluted can increase significantly. This includes activities such as: decanting liquids, sorting materials for recycling, distributing landscaping supplies, etc.

Excessive pollutant loads in stormwater can quickly foul stormwater treatment devices, preventing them from functioning effectively. To avoid this, consider measures that: limit the exposure of potential contaminants to rainfall, how contaminated runoff can be separated, and what additional treatment is needed to avoid contaminants entering the drainage system.



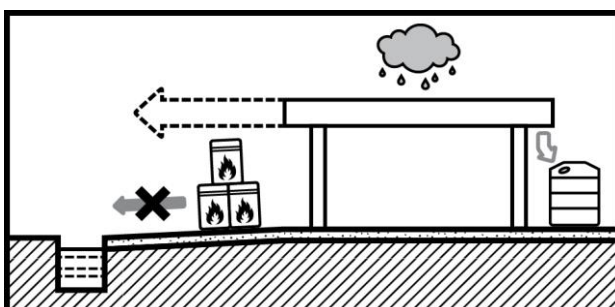
General runoff

- The general pollutants from paved car parking areas is a standard raingarden treatment scenario.
- For areas where litter is likely to be excessive, consider pre-screening stormwater with a gross pollutant trap before it reaches the raingardens.



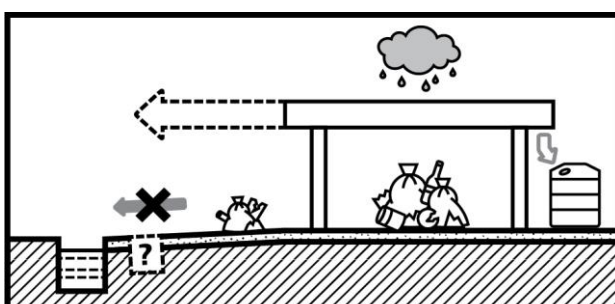
Fuelling areas

- Roofs over fuelling areas minimises the volume of runoff potentially polluted by fuel.
- Providing differential falls in paving each side of a ridge allows runoff that could have fuel contamination to be sent to an interceptor.
- Interceptors separate fuel and oil from stormwater, before the stormwater enters the drainage system.



Liquid contaminants

- Where handling contaminants where spillage can occur, extending roofs over will limit stormwater contamination.
- Providing a perimeter upstand or 'bundling' will aid contain larger spills.
- Refer to Greater Western Water for [trade waste information](#)
- EPA Guidance for [Liquid storage and handling guidelines](#).



Solid contaminants

- Site uses such a distributing landscaping supplies or sorting of materials for recycling can contaminate runoff from paved areas.
- Extend roofs over areas most likely to generate contaminated runoff.
- To avoid high amounts of gross pollutants and particularly sediment from fouling raingardens, consider pre-filtering runoff.

Useful resources

There are a range of government, and non-government organisations that provide information on general WSUD strategies, and more targetted resources for particular technologies.

TOPIC	LINK
General WSUD & stormwater	SDAPP fact sheet: 4.0 Stormwater Management (CASBE) SDAPP fact sheet: 4.1 Site permeability (CASBE) Online Navigator Tool for Victorian planning requirements for stormwater management (Clearwater) Your home technical manual - Stormwater (Australian Government)
Rainwater Tanks	Rainwater tanks (Melbourne Water) Guidance on the use of rainwater tanks (Australian Government) Rainwater (Victoria – Dept. of Health)
Porous Paving	Porous paving raingarden instruction sheet (Melbourne Water) SDAPP fact sheet: 4.1 Site permeability (CASBE)
Swales	Swale raingarden instruction sheet (Melbourne Water)
Planter box raingardens	Planter box raingarden instruction sheet (Melbourne Water) Vegetable raingarden instruction sheet (Melbourne Water)
In-ground raingarden	In-ground raingarden instruction sheet (Melbourne Water) In-ground raingarden with submerged zone (Melbourne Water)
Modelling tools	STORM (Melbourne Water) MUSIC (eWater)
Modelling guidelines	STORM and MUSIC tools (Melbourne Water)
Maintenance	WSUD Maintenance Guidelines (Melbourne Water)
During Construction	Keeping our stormwater clean builders guidelines (EPA & Melbourne Water) How to prevent stormwater pollution on building sites (Victoria - EPA)